Supplementary information for: Natural variation in *Arabidopsis* darkness metabolism tests a carbon starvation model of spaceflight transcriptomes

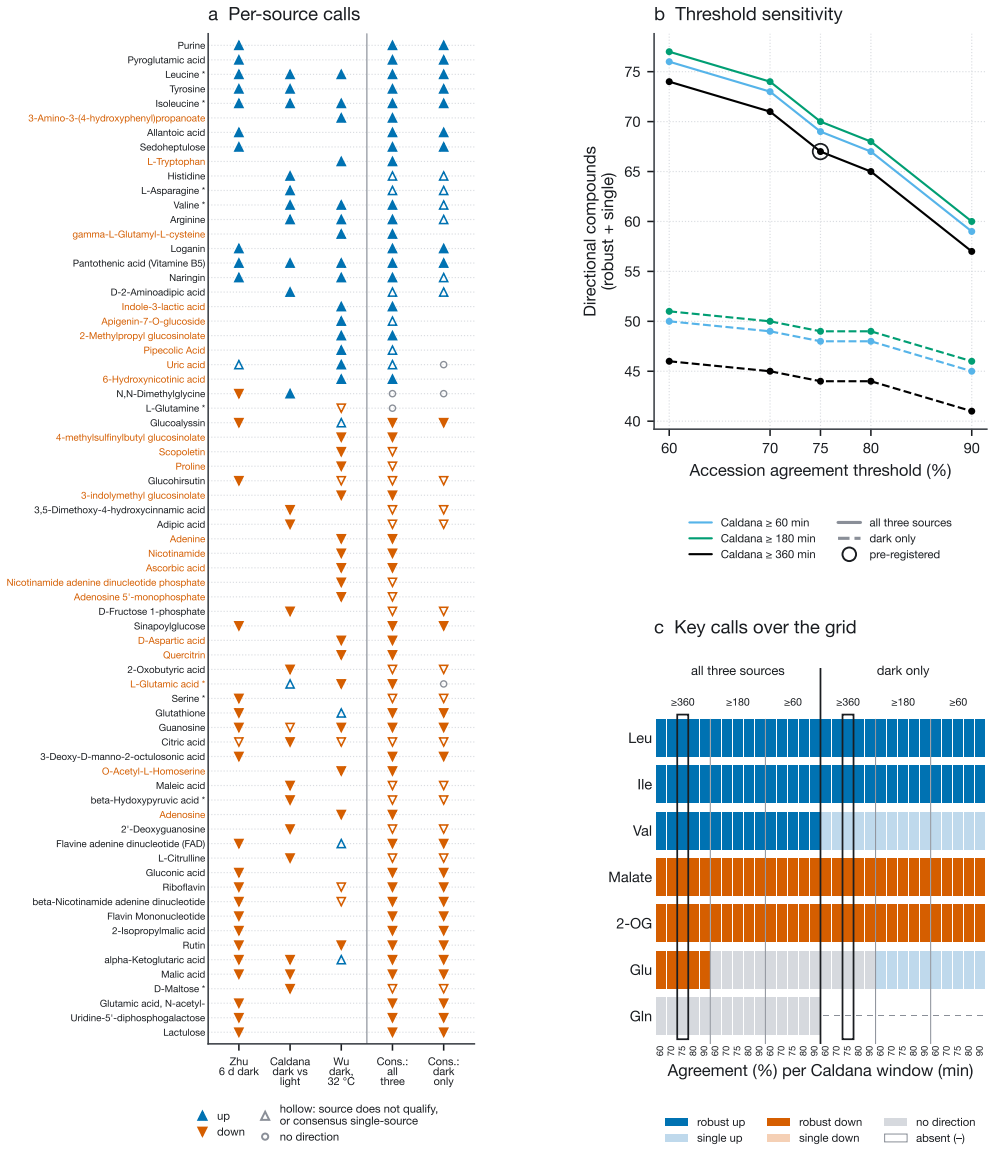
Richard Barker[1*]

[1][TODO — affiliation], [TODO], [TODO].

Corresponding author(s). E-mail(s): dr.richard.barker@gmail.com;

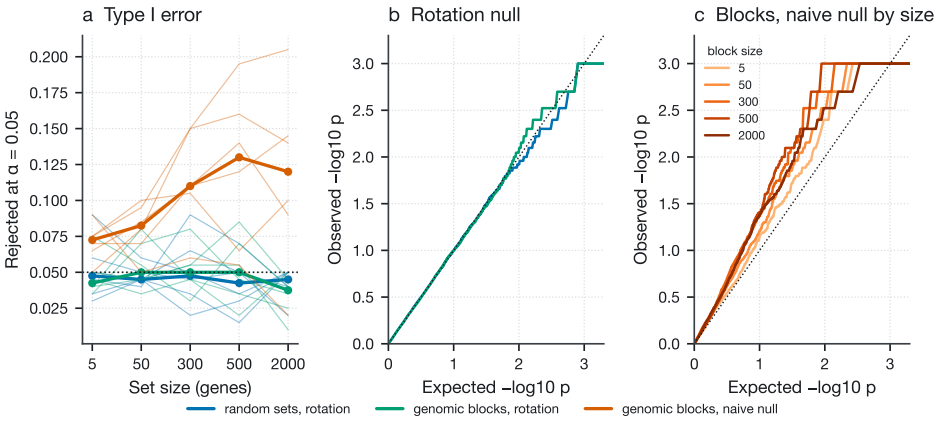
Everything here except Table S1 was added in revision, in answer to a reviewer, and is post hoc: the analysis plan is in `prereg/revision1/` (commit 90ff4b8; hypoxia-signature addendum 8d6a9ae), committed before these analyses ran but after the pre-registered results were known. The pre-registered results (T02, T08, T12 and the T03 verdicts) are unchanged. Each table is generated from the named results table in the repository.

Supplementary figures



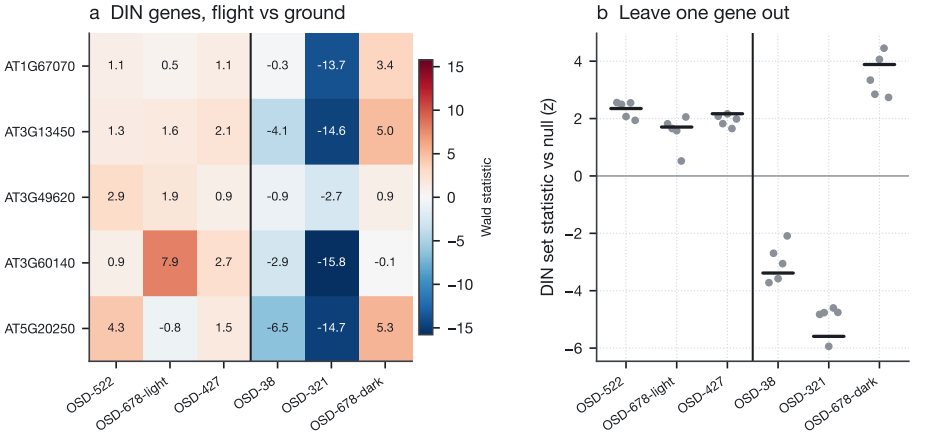
* model compound. Orange names: directional only because of the 32 °C dark+heat source (Wu). Outlined column in c: the pre-registered setting (75 %, ≥ 360 min).

Fig. S1 The darkness reference by source and stratum, and its threshold sensitivity. a, Per-source calls for every compound with a directional or conflicting consensus, or in the model: filled symbols, the source qualifies (or the consensus is robust); hollow, it does not (or the consensus rests on one source). Orange names are directional only because of the 32 °C dark+heat source (Wu). **b**, Directional compounds over accession agreement and Caldana window, for all three sources (solid) and darkness only (dashed). **c**, The key calls over the same grid; the outlined columns are the pre-registered setting.



200 sets per size and type in each of the six flight contrasts; 999 rotation offsets or naive draws per test. Thin lines in a: single contrasts.

Fig. S2 Calibration of the genomic-rotation null. a, Share of 200 random gene sets or contiguous genomic blocks rejected at $\alpha = 0.05$, by set size; thin lines single flight contrasts, thick lines their median. **b**, Quantile–quantile plot of rotation p-values, pooled over contrasts and sizes. **c**, Genomic blocks against a naive random-gene null, by block size: the inflation that gene clustering causes and the rotation null removes.



Left of the vertical line: lit flights; right: dark flights. b: bar = all five genes; dots = one gene left out. OSD-522 uses the 924-labelling sample permutation, the others the rotation null.

Fig. S3 The five DIN genes. a, Wald statistic of each gene in each flight contrast. **b**, The DIN set statistic against its null (z) with all five genes (bar) and with each left out (dots).

Supplementary tables

Table S1: Where each entry of Table 1 and of the hardware and hypoxia designs is stated. Values as given by the source, condensed; "not stated" after searching the full text or deposit. Wu et al. is read in its accepted manuscript (Wageningen University repository, https://edepot.wur.nl/426556; line numbers are the manuscript's).

Source	Field	Value	Where stated
Zhu 2024 [1]	Material	Leaf, 50 mg ground; one plant per accession per timepoint	Methods, LC-MS analysis; Plant material and sample preparation
	Age, growth	35 d after germination; short days in a greenhouse; photoperiod hours, light and temperature not stated	Plant material and sample preparation
	Seasons	Autumn 2018 and spring 2019, two biological replicates	Plant material and sample preparation
	Darkness	0 and 6 d; temperature in darkness not stated	Abstract; whole text searched
	Sampling	Within 30 min at 10:00 (GMT+2)	Plant material and sample preparation
	Platform	Acquity UPLC with Exactive Orbitrap; negative mode, m/z 100–1500	LC-MS analysis
	Values	BLUPs (lme4) over the two seasons; batches standardised to quality controls	Technical Validation; Plant material
	Accessions	259	Plant material and sample preparation
Zhu 2022 [2]	Growth	Short days, 8 h light, 250 μE m^{-2} s^{-1}, 20/16 °C, humidity 60/75 %	Materials and methods, Plant materials and growth conditions
	Sampling	At each harvest, within 30 min at 10:00 (GMT+2)	Same
Wu 2018 [3]	Control	16 h light, 150 μE m^{-2} s^{-1}, 20/16 °C, humidity 60/75 %	Lines 510–512
	Stress	Darkness, 32 °C, humidity 75 %, continuous 1,280 min, matching Caldana et al.'s stress	Lines 512–515
	Age, harvest	Control harvested at 42 d after germination, from 4 h into the light period; stress plants after 1,280 min	Lines 515–518
	Material	About 100 mg rosettes; three plants pooled per biological replicate	Lines 518–520, 548–550
	Platform	Acquity UPLC with Exactive; positive and negative modes; day- and sample-median normalisation	Lines 558–570, 594–595
	Filter	Traits present in at least 50 % of accessions	Lines 621–622
	Accessions	309 (314 less five with poor germination)	Lines 604–606
	Time course	Material of Caldana et al. 2011, Col-0, re-profiled by LC–MS; conditions 21-D and 21-L (21 °C, 150 μE) among eight	Lines 176–186, 475–484

Source	Field	Value	Where stated
	Column labels	Supplementary Table 7 labels conditions 4/20/30 (text: 4/21/32 °C); the label key is not stated	Supplementary Table 7; text and Supplementary Information searched
Caldana 2011 [4]	Conditions	Grown at 21 °C, 150 μE; kept there or moved to seven environments including 21 °C darkness; 0 and 22 time points	Abstract (full text not accessible to us)
E-MTAB-375 [5]	Material	Columbia-0 (stock CS906), wild type, rosette leaf	Sample and data relationship file
OSD-38 [6]	Comparator	22 plates in sealed PDFUs in BRIC canisters at Kennedy Space Center; 22 plates in foil inside opaque, air-tight plastic containers at Ohio University; 4 d cold, then 76 h growth; continuous dark	OSDR growth protocol
	Fixation	RNAlater, 12 h, then −80 °C, both arms	OSDR sample collection protocol
OSD-136 [7, 8]	Design	Wassilewskija roots, 5 d old; 24 h at 97 kPa, 50 kPa (O_2 10 kPa) or 50 kPa with O_2 restored to 21 kPa; three replicates	OSDR treatment protocol; Zhou et al., Abstract, Methods, Table 1
GSE14502 [9, 10]	Design	7-d Col-0 seedlings (p35S:HF-RPL18), vertical MS plates, 23 °C; 2 h argon from the end of the light period; roots and shoots; total mRNA from the 35S line; ATH1 arrays, RMA	GEO sample records; Mustroph et al., Materials and Methods

Table S2: **Reference consensus by stratum (T14).** Per source: effect ($\log_2$; * the source qualifies) and support (Zhu, Wu: share of accessions moving with the median, accessions in brackets; Caldana: Welch p, values in brackets). Consensus with all three sources (= T02) and with darkness only (Zhu, Caldana); "dark+heat only": directional only because of Wu's 32 °C stress.

Compound	KEGG	Zhu	Caldana	Wu (32 °C)	All three	Dark only	Heat only
N,N-Dimethylglycine	C01026	−0.59* / 100 % (259)	0.93* / 1.6×10^{-4} (18)	–	confl.	confl.	
L-Glutamine	C00064	–	–	−0.02 / 50 % (301)	none	–	
Methionine	C00073	–	–	0.12 / 59 % (301)	none	–	
Choline	C00114	–	0.01 / 0.946 (18)	–	none	none	
5'-Deoxy-5'-(methylthio)adenos	C00170	–	−0.05 / 0.580 (18)	–	none	none	
cis-Aconitic acid	C00417	–	–	0.14 / 57 % (301)	none	–	
Cytidine	C00475	–	−0.09 / 0.611 (18)	–	none	none	
5-Hydroxy-L-tryptophan	C00643	–	−0.09 / 0.381 (17)	–	none	none	
p-Coumaric acid	C00811	–	−0.14 / 0.450 (18)	–	none	none	
Saccharic acid	C00818	−0.10 / 68 % (259)	–	–	none	none	
4-Guanidinobutyric acid	C01035	–	0.00 / 0.996 (18)	–	none	none	
Allantoin	C01551	–	0.46 / 0.146 (18)	–	none	none	
Maltotriose	C01835	–	−0.33 / 0.273 (18)	–	none	none	
Benzyl glucosinolate	C02153	–	–	0.36 / 73 % (300)	none	–	
alpha-L-Fucose 1-phosphate	C02985	–	−0.02 / 0.922 (18)	–	none	none	
5-O-(Indol-3-ylacetyl-myo-inos	C04695	–	–	−0.57 / 75 % (295)	none	–	
Allothreonine	C05519	–	−0.36 / 0.068 (18)	–	none	none	
N-Acetyl-D-phenylalanine	C05620	–	–	−0.22 / 65 % (301)	none	–	
Nicotinate D-ribonucleoside	C05841	–	–	−0.09 / 58 % (285)	none	–	
Hydroxymethylphosphonic acid	C06455	–	0.06 / 0.664 (14)	–	none	none	
5-methylthiopentylglucosinolat	C08401	–	–	0.51 / 63 % (298)	none	–	
4-pentenylglucosinolate	C08403	–	–	0.16 / 53 % (192)	none	–	
4-methylthiobutyl glucosinolat	C08409	–	–	0.01 / 50 % (264)	none	–	
3-methylsulfinylpropyl-glucosi	C08411	–	–	−0.96 / 73 % (199)	none	–	
3-methylthiopropyl-glucosinola	C08412	–	–	−0.21 / 57 % (269)	none	–	
3-butenylglucosinolate	C08415	–	–	−0.04 / 51 % (301)	none	–	
2-Picolinic acid	C10164	–	–	0.22 / 72 % (301)	none	–	
7-methylthioheptyl glucosinola	C17252	–	–	0.25 / 63 % (270)	none	–	
8-Methylthiooctyl glucosinolat	C17254	–	–	0.01 / 51 % (301)	none	–	
Purine	C15587	5.11* / 100 % (259)	–	–	robust ↑	robust ↑	
Pyroglutamic acid	C01879	1.43* / 100 % (259)	–	–	robust ↑	robust ↑	
Leucine	C00123	4.92* / 100 % (259)	2.40* / 2.9×10^{-8} (18)	4.74* / 100 % (301)	robust ↑	robust ↑	
Tyrosine	C00082	5.06* / 100 % (259)	2.36* / 1.8×10^{-6} (18)	–	robust ↑	robust ↑	

(continued)

Compound	KEGG	Zhu	Caldana	Wu (32 °C)	All three	Dark only	Heat only
Isoleucine	C00407	3.15* / 100 % (259)	2.17* / 2.7×10^{-7} (18)	4.42* / 100 % (301)	robust ↑	robust ↑	
3-Amino-3-(4-hydroxyphenyl)pro	C04368	–	–	5.05* / 100 % (301)	robust ↑	–	yes
Allantoic acid	C00499	4.01* / 100 % (259)	–	–	robust ↑	robust ↑	
Sedoheptulose	C02076	1.46* / 100 % (259)	–	–	robust ↑	robust ↑	
L-Tryptophan	C00078	–	–	5.41* / 100 % (301)	robust ↑	–	yes
Valine	C00183	–	0.90* / 4.3×10^{-5} (18)	2.49* / 100 % (301)	robust ↑	single ↑	
Arginine	C00062	–	2.58* / 1.8×10^{-6} (18)	2.34* / 100 % (301)	robust ↑	single ↑	
gamma-L-Glutamyl-L-cysteine	C00669	–	–	2.39* / 100 % (301)	robust ↑	–	yes
Loganin	C01433	2.28* / 100 % (259)	–	–	robust ↑	robust ↑	
Pantothenic acid (Vitamine B5)	C00864	1.26* / 100 % (259)	0.25* / 0.019 (18)	2.02* / 100 % (301)	robust ↑	robust ↑	
Naringin	C09789	1.42* / 88 % (259)	–	0.36* / 81 % (298)	robust ↑	single ↑	
Indole-3-lactic acid	C02043	–	–	1.00* / 94 % (301)	robust ↑	–	yes
2-Methylpropyl glucosinolate	C17256	–	–	1.26* / 93 % (284)	robust ↑	–	yes
6-Hydroxynicotinic acid	C01020	–	–	0.50* / 92 % (293)	robust ↑	–	yes
Glucoalyssin	C08400	−0.71* / 96 % (259)	–	0.14 / 54 % (301)	robust ↓	robust ↓	
4-methylsulfinylbutyl glucosin	C08419	–	–	−1.19* / 96 % (199)	robust ↓	–	yes
3-indolymethyl glucosinolate	C05837	–	–	−0.51* / 90 % (301)	robust ↓	–	yes
Adenine	C00147	–	–	−0.90* / 93 % (301)	robust ↓	–	yes
Nicotinamide	C00153	–	–	−1.32* / 92 % (301)	robust ↓	–	yes
Ascorbic acid	C00072	–	–	−1.61* / 92 % (274)	robust ↓	–	yes
Sinapoylglucose	C01175	−0.73* / 96 % (259)	–	–	robust ↓	robust ↓	
D-Aspartic acid	C00049	–	–	−0.43* / 92 % (301)	robust ↓	–	yes
Quercitrin	C01750	–	–	−1.97* / 99 % (298)	robust ↓	–	yes
L-Glutamic acid	C00025	–	0.08 / 0.337 (18)	−0.51* / 97 % (301)	robust ↓	none	yes
Glutathione	C00051	−0.47* / 98 % (259)	–	0.04 / 54 % (301)	robust ↓	robust ↓	
Guanosine	C00387	−1.06* / 99 % (259)	−0.36 / 0.062 (18)	−0.93* / 90 % (301)	robust ↓	robust ↓	
3-Deoxy-D-manno-2-octulosonic	C01187	−0.51* / 100 % (259)	–	–	robust ↓	robust ↓	
O-Acetyl-L-Homoserine	C01077	–	–	−1.45* / 100 % (301)	robust ↓	–	yes
Adenosine	C00212	–	–	−1.32* / 97 % (301)	robust ↓	–	yes
Flavine adenine dinucleotide (	C00016	−0.80* / 100 % (259)	–	0.28 / 69 % (300)	robust ↓	robust ↓	
Gluconic acid	C00257	−0.38* / 100 % (259)	–	–	robust ↓	robust ↓	
Riboflavin	C00255	−1.11* / 100 % (259)	–	−0.16 / 65 % (301)	robust ↓	robust ↓	
beta-Nicotinamide adenine dinu	C00003	−0.61* / 100 % (259)	–	−0.00 / 50 % (301)	robust ↓	robust ↓	
Flavin Mononucleotide	C00061	−0.56* / 100 % (259)	–	–	robust ↓	robust ↓	
2-Isopropylmalic acid	C02504	−0.73* / 98 % (259)	–	–	robust ↓	robust ↓	
Rutin	C05625	−0.99* / 99 % (259)	–	−0.63* / 81 % (301)	robust ↓	robust ↓	
alpha-Ketoglutaric acid	C00026	−1.68* / 100 % (259)	−0.72* / 0.003 (18)	0.12 / 59 % (301)	robust ↓	robust ↓	
Malic acid	C00149	−1.19* / 100 % (259)	−0.58* / 7.8×10^{-4} (18)	–	robust ↓	robust ↓	
Glutamic acid, N-acetyl-	C00624	−2.79* / 100 % (259)	–	–	robust ↓	robust ↓	
Uridine-5'-diphosphogalactose	C00052	−1.91* / 100 % (259)	–	–	robust ↓	robust ↓	
Lactulose	C07064	−4.13* / 100 % (259)	–	–	robust ↓	robust ↓	
Histidine	C00135	–	1.50* / 4.0×10^{-5} (18)	–	single ↑	single ↑	
L-Asparagine	C00152	–	1.51* / 1.4×10^{-4} (18)	–	single ↑	single ↑	
D-2-Aminoadipic acid	C00956	–	0.79* / 0.021 (18)	–	single ↑	single ↑	
Apigenin-7-O-glucoside	C04608	–	–	0.42* / 82 % (301)	single ↑	–	yes
Pipecolic Acid	C00408	–	–	1.18* / 82 % (297)	single ↑	–	yes
Uric acid	C00366	0.25 / 74 % (259)	–	0.56* / 82 % (301)	single ↑	none	yes
Scopoletin	C01752	–	–	−0.44* / 80 % (301)	single ↓	–	yes
Proline	C00148	–	–	−0.48* / 75 % (301)	single ↓	–	yes
Glucohirsutin	C17271	−0.26* / 82 % (259)	–	−0.10 / 59 % (301)	single ↓	single ↓	
3,5-Dimethoxy-4-hydroxycinnami	C00482	–	−0.53* / 0.022 (15)	–	single ↓	single ↓	
Adipic acid	C06104	–	−0.18* / 0.015 (18)	–	single ↓	single ↓	
Nicotinamide adenine dinucleot	C00006	–	–	−1.39* / 89 % (275)	single ↓	–	yes
Adenosine 5'-monophosphate	C00020	–	–	−0.52* / 88 % (301)	single ↓	–	yes
D-Fructose 1-phosphate	C01094	–	−0.83* / 0.030 (17)	–	single ↓	single ↓	
2-Oxobutyric acid	C00109	–	−0.25* / 0.035 (18)	–	single ↓	single ↓	
Serine	C00065	−0.29* / 83 % (259)	–	–	single ↓	single ↓	
Citric acid	C00158	−0.12 / 74 % (259)	−0.73* / 0.006 (18)	−0.03 / 50 % (301)	single ↓	single ↓	
Maleic acid	C01384	–	−0.75* / 7.7×10^{-4} (18)	–	single ↓	single ↓	
beta-Hydoxypyruvic acid	C00168	–	−0.73* / 0.003 (18)	–	single ↓	single ↓	
2'-Deoxyguanosine	C00330	–	−0.91* / 0.001 (18)	–	single ↓	single ↓	
L-Citrulline	C00327	–	−2.46* / 7.3×10^{-6} (18)	–	single ↓	single ↓	
D-Maltose	C00208	–	−1.23* / 9.8×10^{-4} (18)	–	single ↓	single ↓	

Table S3: **Threshold sensitivity of the reference (T15).** Accession agreement (%) and the start of the Caldana window (min). Key calls: robust or single, with direction. Model: agrees/disagrees among the scorable predictions; last column, robust responses the model did not predict.

Stratum	Agr.	Cald.	Rob.	Sing.	Leu	Ile	Val	Malate	2-OG	Glu	Gln	Model	Not pred.
all three	60	60	45	31	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	conflicting	none	5/0	42
dark only	60	60	27	23	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	single ↑	–	6/0	25
all three	60	180	45	32	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	conflicting	none	5/0	42
dark only	60	180	27	24	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	single ↑	–	6/0	25
all three	60	360	46	28	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	robust ↓	none	5/1	42
dark only	60	360	27	19	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	none	–	5/0	25
all three	70	60	46	27	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	conflicting	none	5/0	43
dark only	70	60	27	22	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	single ↑	–	6/0	25
all three	70	180	46	28	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	conflicting	none	5/0	43
dark only	70	180	27	23	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	single ↑	–	6/0	25
all three	70	360	47	24	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	robust ↓	none	5/1	43

(continued)

Stratum	Agr.	Cald.	Rob.	Sing.	Leu	Ile	Val	Malate	2-OG	Glu	Gln	Model	Not pred.
dark only	70	360	27	18	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	none	–	5/0	25
all three	75	60	44	25	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	conflicting	none	5/0	41
dark only	75	60	26	22	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	single ↑	–	6/0	24
all three	75	180	44	26	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	conflicting	none	5/0	41
dark only	75	180	26	23	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	single ↑	–	6/0	24
all three (pre-reg.)	75	360	45	22	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	robust ↓	none	5/1	41
dark only (pre-reg.)	75	360	26	18	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	none	–	5/0	24
all three	80	60	44	23	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	conflicting	none	5/0	41
dark only	80	60	26	22	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	single ↑	–	6/0	24
all three	80	180	44	24	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	conflicting	none	5/0	41
dark only	80	180	26	23	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	single ↑	–	6/0	24
all three	80	360	45	20	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	robust ↓	none	5/1	41
dark only	80	360	26	18	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	none	–	5/0	24
all three	90	60	43	16	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	conflicting	none	5/0	40
dark only	90	60	26	19	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	single ↑	–	6/0	24
all three	90	180	43	17	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	conflicting	none	5/0	40
dark only	90	180	26	20	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	single ↑	–	6/0	24
all three	90	360	44	13	robust ↑	robust ↑	robust ↑	robust ↓	robust ↓	robust ↓	none	5/1	40
dark only	90	360	26	15	robust ↑	robust ↑	single ↑	robust ↓	robust ↓	none	–	5/0	24

Table S4: How measured names reached KEGG (T16). Candidates from KEGG `find` on each name; the rule takes an exact synonym match; overrides are reviewed by hand (`data/reference/kegg_overrides.tsv`). Wu classes are Wu et al.'s.

Section	Item	n	of
names	measured names sent to KEGG (Zhu and Wu)	112	
names	from wu_stress	70	112
names	from zhu	42	112
KEGG find candidates	0 candidate(s)	48	112
KEGG find candidates	1 candidate(s)	21	112
KEGG find candidates	>1 candidate(s)	43	112
rule match (exact synonym)	exact	56	112
rule match (exact synonym)	none	50	112
rule match (exact synonym)	exact (several hits, lowest ID kept)	4	112
rule match (exact synonym)	D-prefix	2	112
rule match (exact synonym)	mapped by the rule	62	112
reviewed overrides	rows in data/reference/kegg_overrides.tsv	24	
reviewed overrides	rescued (rule found no exact synonym)	16	24
reviewed overrides	corrected (rule picked another entry)	6	24
reviewed overrides	Caldana peak only: Wu's given ID corrected	1	24
reviewed overrides	Caldana peak only: excluded	1	24
after overrides	measured names with a KEGG ID	78	112
after overrides	from wu_stress	45	70
after overrides	from zhu	33	42
after overrides	Caldana peaks (KEGG IDs given by Wu et al.)	35	36
after overrides	Caldana peaks whose given ID an override changed or removed	2	36
Wu class (after overrides)	Amine	1	2
Wu class (after overrides)	Amino acid	10	10
Wu class (after overrides)	Amino acid derivative	1	1
Wu class (after overrides)	Cofactor	1	1
Wu class (after overrides)	Dipeptide	1	1
Wu class (after overrides)	Flavonoid	1	6
Wu class (after overrides)	Glucoside	0	3
Wu class (after overrides)	Glucosinolate	14	27
Wu class (after overrides)	Nucleobase	1	1
Wu class (after overrides)	Nucleoside	1	1
Wu class (after overrides)	Nucleoside derivative	1	1
Wu class (after overrides)	Nucleotide	2	2
Wu class (after overrides)	Organic acid	7	7

(continued)

Section	Item	n	of
Wu class (after overrides)	Phenylpropanoid	2	5
Wu class (after overrides)	Sugar	1	1
Wu class (after overrides)	Vitamin	1	1

Table S5: **All 36 pre-registered gene-set tests (T08).** Set statistic: mean gene statistic (signed) or mean absolute statistic (unsigned) in the set minus the genome; z and p against the null; BH q over the 36. n: measured/listed members. Members: `prereg/gene_sets.tsv` (KEGG retrieved 30 October 2026).

Set	Contrast	Stat.	Null	Pred.	Obs.	Verdict	Set stat.	z	p	q	n
DIN	OSD-522	signed	permutation	up	none	MISMATCH	2.028	2.35	0.028	0.127	5/5
BCAA degradation	OSD-522	signed	permutation	up	none	MISMATCH	−0.194	−0.75	0.452	0.562	49/53
Wu control-only	OSD-522	unsigned	permutation	none	none	MATCH	−0.019	−0.12	0.506	0.608	208/301
Wu stress-only	OSD-522	unsigned	permutation	enriched	none	MISMATCH	0.064	1.25	0.108	0.260	367/491
Wu shared	OSD-522	unsigned	permutation	none	none	MATCH	0.035	1.06	0.145	0.307	1329/1954
Wu darkness metabolites	OSD-522	unsigned	permutation	enriched	none	MISMATCH	0.052	2.06	0.028	0.127	350/540
DIN	OSD-678-light	signed	rotation	up	none	MISMATCH	2.199	1.70	0.089	0.229	5/5
BCAA degradation	OSD-678-light	signed	rotation	up	none	MISMATCH	0.797	2.05	0.040	0.143	53/53
Wu control-only	OSD-678-light	unsigned	rotation	none	none	MATCH	0.032	0.23	0.403	0.557	233/301
Wu stress-only	OSD-678-light	unsigned	rotation	enriched	none	MISMATCH	−0.056	−0.57	0.716	0.781	423/491
Wu shared	OSD-678-light	unsigned	rotation	none	enriched	MISMATCH	0.147	2.85	0.002	0.011	1500/1954
Wu darkness metabolites	OSD-678-light	unsigned	rotation	enriched	none	MISMATCH	0.125	1.19	0.120	0.270	400/540
DIN	OSD-427	signed	rotation	up	none	MISMATCH	1.731	2.17	0.032	0.127	5/5
BCAA degradation	OSD-427	signed	rotation	up	none	MISMATCH	−0.070	−0.29	0.780	0.802	52/53
Wu control-only	OSD-427	unsigned	rotation	none	none	MATCH	0.138	1.68	0.052	0.161	237/301
Wu stress-only	OSD-427	unsigned	rotation	enriched	none	MISMATCH	0.011	0.17	0.423	0.562	430/491
Wu shared	OSD-427	unsigned	rotation	none	none	MATCH	0.046	1.60	0.054	0.161	1533/1954
Wu darkness metabolites	OSD-427	unsigned	rotation	enriched	none	MISMATCH	0.044	0.72	0.239	0.388	408/540
DIN	OSD-38	signed	rotation	none	down	MISMATCH	−2.879	−3.38	0.002	0.011	5/5
BCAA degradation	OSD-38	signed	rotation	none	none	MATCH	−0.451	−1.71	0.081	0.224	53/53
Wu control-only	OSD-38	unsigned	rotation	none	none	MATCH	−0.033	−0.37	0.636	0.739	254/301
Wu stress-only	OSD-38	unsigned	rotation	none	none	MATCH	0.010	0.15	0.440	0.562	452/491
Wu shared	OSD-38	unsigned	rotation	none	none	MATCH	0.017	0.48	0.319	0.479	1639/1954
Wu darkness metabolites	OSD-38	unsigned	rotation	none	none	MATCH	0.060	0.87	0.195	0.353	447/540
DIN	OSD-321	signed	rotation	none	down	MISMATCH	−12.450	−5.59	1.0×10^{-4}	0.002	5/5
BCAA degradation	OSD-321	signed	rotation	none	down	MISMATCH	−4.909	−6.88	1.0×10^{-4}	0.002	52/53
Wu control-only	OSD-321	unsigned	rotation	none	none	MATCH	0.074	0.32	0.371	0.534	241/301
Wu stress-only	OSD-321	unsigned	rotation	none	none	MATCH	0.153	0.85	0.196	0.353	426/491
Wu shared	OSD-321	unsigned	rotation	none	none	MATCH	0.079	0.79	0.227	0.388	1505/1954
Wu darkness metabolites	OSD-321	unsigned	rotation	none	none	MATCH	0.181	0.88	0.194	0.353	399/540
DIN	OSD-678-dark	signed	rotation	none	up	MISMATCH	2.867	3.88	4.0×10^{-4}	0.005	5/5
BCAA degradation	OSD-678-dark	signed	rotation	none	up	MISMATCH	0.773	3.42	1.0×10^{-3}	0.009	53/53
Wu control-only	OSD-678-dark	unsigned	rotation	none	none	MATCH	−0.140	−1.91	0.978	0.978	233/301
Wu stress-only	OSD-678-dark	unsigned	rotation	none	none	MATCH	−0.031	−0.57	0.715	0.781	423/491
Wu shared	OSD-678-dark	unsigned	rotation	none	none	MATCH	0.020	0.70	0.248	0.388	1500/1954
Wu darkness metabolites	OSD-678-dark	unsigned	rotation	none	none	MATCH	−0.038	−0.63	0.741	0.784	400/540

Table S6: **Calibration of the rotation null (T20).** Share of 200 sets rejected at $\alpha = 0.05$, median and range over the six flight contrasts. A calibrated null rejects about 0.05.

Set type	Null	Size	Median	Range
contiguous genomic blocks	naive	5	0.073	0.050–0.090
contiguous genomic blocks	naive	50	0.083	0.050–0.100
contiguous genomic blocks	naive	300	0.110	0.060–0.150
contiguous genomic blocks	naive	500	0.130	0.055–0.195
contiguous genomic blocks	naive	2000	0.120	0.020–0.205
contiguous genomic blocks	rotation	5	0.042	0.035–0.075
contiguous genomic blocks	rotation	50	0.050	0.035–0.080
contiguous genomic blocks	rotation	300	0.050	0.030–0.080
contiguous genomic blocks	rotation	500	0.050	0.020–0.085
contiguous genomic blocks	rotation	2000	0.038	0.010–0.050
random genes	naive	5	0.058	0.030–0.095
random genes	naive	50	0.050	0.025–0.065
random genes	naive	300	0.050	0.020–0.090

(continued)

Set type	Null	Size	Median	Range
random genes	naive	500	0.045	0.015–0.065
random genes	naive	2000	0.048	0.025–0.055
random genes	rotation	5	0.048	0.030–0.090
random genes	rotation	50	0.045	0.040–0.060
random genes	rotation	300	0.048	0.020–0.090
random genes	rotation	500	0.043	0.015–0.070
random genes	rotation	2000	0.045	0.020–0.050

Table S7: **Exact sample permutation for the 3-versus-3 contrasts (T21).** All 20 labellings, observed included, so $p \geq 0.1$. Rotation p and outcome as pre-registered (T08). Last column: OSD-38 sets on PyDESeq2 statistics from GeneLab's raw counts (Spearman with GeneLab's 0.988 over 27,309 genes).

Set	Contrast	Stat.	Perm. p	Rotation p	T08 outcome	PyDESeq2 p
DIN	OSD-38	signed	0.1	0.0015	down	5.0×10^{-4}
BCAA degradation	OSD-38	signed	0.3	0.0808	none	0.2137
Wu control-only	OSD-38	unsigned	0.2	0.6361	none	0.5251
Wu stress-only	OSD-38	unsigned	0.1	0.4396	none	0.2086
Wu shared	OSD-38	unsigned	0.1	0.3193	none	0.2251
Wu darkness metabolites	OSD-38	unsigned	0.1	0.1946	none	0.2263
DIN	OSD-678-light	signed	0.1	0.0890	none	
BCAA degradation	OSD-678-light	signed	0.2	0.0397	none	
Wu control-only	OSD-678-light	unsigned	1.0	0.4026	none	
Wu stress-only	OSD-678-light	unsigned	0.1	0.7163	none	
Wu shared	OSD-678-light	unsigned	0.1	0.0019	enriched	
Wu darkness metabolites	OSD-678-light	unsigned	0.4	0.1201	none	
DIN	OSD-678-dark	signed	0.4	4.0×10^{-4}	up	
BCAA degradation	OSD-678-dark	signed	0.3	1.0×10^{-3}	up	
Wu control-only	OSD-678-dark	unsigned	1.0	0.9776	none	
Wu stress-only	OSD-678-dark	unsigned	1.0	0.7150	none	
Wu shared	OSD-678-dark	unsigned	0.8	0.2477	none	
Wu darkness metabolites	OSD-678-dark	unsigned	1.0	0.7408	none	

Table S8: **The five DIN genes (T22).** Top: Wald statistic per contrast; then the set p (OSD-522 sample permutation, others rotation) with all five genes and with each left out. Bottom: $\log_2$(normalised count + 1) per sample in the 3-versus-3 contrasts, three flight then three ground (OSDR sample order). In OSD-678 dark, flight replicate 3 carries the rise.

	OSD-522	OSD-678-light	OSD-427	OSD-38	OSD-321	OSD-678-dark
AT1G67070 (Wald)	1.1	0.5	1.1	−0.3	−13.7	3.4
AT3G13450 (Wald)	1.3	1.6	2.1	−4.1	−14.6	5.0
AT3G49620 (Wald)	2.9	1.9	0.9	−0.9	−2.7	0.9
AT3G60140 (Wald)	0.9	7.9	2.7	−2.9	−15.8	−0.1
AT5G20250 (Wald)	4.3	−0.8	1.5	−6.5	−14.7	5.3
all five: p	0.0281	0.0890	0.0318	0.0015	1.0×10^{-4}	4.0×10^{-4}
without AT1G67070: p	0.0152	0.0713	0.0368	4.0×10^{-4}	1.0×10^{-4}	0.0021
without AT3G13450: p	0.0238	0.0946	0.0680	0.0103	2.0×10^{-4}	0.0077
without AT3G49620: p	0.0476	0.1128	0.0318	0.0011	1.0×10^{-4}	1.0×10^{-4}
without AT3G60140: p	0.0238	0.5605	0.0967	0.0049	1.0×10^{-4}	2.0×10^{-4}
without AT5G20250: p	0.0584	0.0454	0.0476	0.0400	1.0×10^{-4}	0.0092
OSD-38 AT1G67070	7.37	7.35	7.50	7.48	7.67	7.34
OSD-38 AT3G13450	8.06	8.59	8.45	8.97	9.47	9.15
OSD-38 AT3G49620	3.37	3.90	3.08	4.88	4.41	4.56
OSD-38 AT3G60140	5.49	6.92	7.15	7.15	9.26	8.48
OSD-38 AT5G20250	12.41	12.62	12.44	14.21	14.02	14.05
OSD-678-light AT1G67070	9.97	9.17	7.61	9.12	8.69	8.82
OSD-678-light AT3G13450	13.90	13.60	12.35	12.60	12.70	12.75

(continued)

	OSD-522	OSD-678-light	OSD-427	OSD-38	OSD-321	OSD-678-dark
OSD-678-light AT3G49620	7.89	8.36	8.11	6.13	6.85	7.43
OSD-678-light AT3G60140	13.17	12.97	11.37	5.66	7.49	7.66
OSD-678-light AT5G20250	16.66	16.55	15.42	16.74	16.74	16.43
OSD-678-dark AT1G67070	7.16	8.23	10.83	8.04	7.07	7.27
OSD-678-dark AT3G13450	10.73	12.46	14.85	11.32	11.40	10.99
OSD-678-dark AT3G49620	9.12	9.60	6.56	7.62	8.27	8.79
OSD-678-dark AT3G60140	10.06	11.34	12.98	12.23	12.12	11.58
OSD-678-dark AT5G20250	13.89	15.32	17.62	14.25	14.17	13.71

Table S9: **OSD-678 flight × light (T19).** One edgeR-TMM/limma-voom model of the twelve Col-0 samples; rotation tests on the moderated t of the interaction (primary; BH over eleven sets) and of flight in light and in darkness (context). Marginal limma t against GeneLab's Wald: Spearman 0.97 (light), 0.94 (dark).

Set	Term	Stat.	z	p	q	Outcome	n
DIN	interaction	signed	8.5×10^{-4}	0.9808	0.9808	none	5
BCAA degradation	interaction	signed	0.1	0.8983	0.9808	none	52
Photosystem	interaction	signed	−11.5	1.0×10^{-4}	5.5×10^{-4}	down	31
Fermentation	interaction	signed	1.6	0.1015	0.1861	none	6
Hypoxia-responsive	interaction	signed	0.4	0.6453	0.8873	none	3
Glyoxylate/dicarboxylate	interaction	signed	−4.3	2.0×10^{-4}	7.3×10^{-4}	down	77
Carbon fixation	interaction	signed	−5.6	1.0×10^{-4}	5.5×10^{-4}	down	69
Starch and sucrose	interaction	signed	−2.4	0.0175	0.0385	down	154
Glycolysis	interaction	signed	−2.9	0.0043	0.0118	down	115
Wu stress-only	interaction	unsigned	0.0	0.4737	0.7443	none	404
Wu darkness metabolites	interaction	unsigned	−0.6	0.7356	0.8991	none	388
DIN	flight light	signed	1.1	0.2913	0.3204	none	5
BCAA degradation	flight light	signed	1.6	0.1063	0.1949	none	52
Photosystem	flight light	signed	−6.5	1.0×10^{-4}	5.5×10^{-4}	down	31
Fermentation	flight light	signed	1.1	0.2691	0.3204	none	6
Hypoxia-responsive	flight light	signed	0.4	0.6905	0.6905	none	3
Glyoxylate/dicarboxylate	flight light	signed	−2.8	0.0052	0.0143	down	77
Carbon fixation	flight light	signed	−4.8	1.0×10^{-4}	5.5×10^{-4}	down	69
Starch and sucrose	flight light	signed	−3.0	0.0027	0.0099	down	154
Glycolysis	flight light	signed	−2.2	0.0296	0.0651	none	115
Wu stress-only	flight light	unsigned	0.6	0.2800	0.3204	none	404
Wu darkness metabolites	flight light	unsigned	0.7	0.2195	0.3204	none	388
DIN	flight dark	signed	2.2	0.0328	0.0722	none	5
BCAA degradation	flight dark	signed	2.6	0.0118	0.0324	up	52
Photosystem	flight dark	signed	12.5	1.0×10^{-4}	5.5×10^{-4}	up	31
Fermentation	flight dark	signed	−1.7	0.0834	0.1310	none	6
Hypoxia-responsive	flight dark	signed	−0.2	0.8191	0.9510	none	3
Glyoxylate/dicarboxylate	flight dark	signed	4.2	1.0×10^{-4}	5.5×10^{-4}	up	77
Carbon fixation	flight dark	signed	3.0	0.0039	0.0143	up	69
Starch and sucrose	flight dark	signed	−0.4	0.6487	0.8920	none	154
Glycolysis	flight dark	signed	1.9	0.0548	0.1005	none	115
Wu stress-only	flight dark	unsigned	−1.4	0.9213	0.9510	none	404
Wu darkness metabolites	flight dark	unsigned	−1.7	0.9510	0.9510	none	388

Table S10: **PaintOmics control jobs (T17).** E: job A without metabolites; F: A with effect-scaled metabolite values; G, H: A with genes or proteins only (plus the layer); I, J: the same for D; K1, K2: B with its relevant genes cut at random to C's number. Also the per-omic p in job A and the relevant compounds without a hub result.

Job	vs	Quantity	Value
E K3ju047ZWB	A	pathways at BH q < 0.05 (job; parent)	6; 6
E K3ju047ZWB	A	pathways gained	none
E K3ju047ZWB	A	pathways lost	none
E K3ju047ZWB	A	Spearman rho of combined p	0.9518
E K3ju047ZWB	A	hub compounds tested (job; parent)	0; 32
E K3ju047ZWB	A	hubs at FDR < 0.05 (job; parent)	0; 11
E K3ju047ZWB	A	hubs gained	none
E K3ju047ZWB	A	hubs lost	(S)-Malate; 2-Isopropylmalate; 2-Oxoglutarate; Glucobrassicin; Glutathione; L-Ascorbate; L-Glutamate; L-Isoleucine; L-Leucine; L-Valine; gamma-L-Glutamyl-L-cysteine
F L46ds55AiY	A	pathways at BH q < 0.05 (job; parent)	6; 6
F L46ds55AiY	A	pathways gained	none
F L46ds55AiY	A	pathways lost	none
F L46ds55AiY	A	Spearman rho of combined p	1.0
F L46ds55AiY	A	hub compounds tested (job; parent)	32; 32
F L46ds55AiY	A	hubs at FDR < 0.05 (job; parent)	11; 11
F L46ds55AiY	A	hubs gained	none
F L46ds55AiY	A	hubs lost	none
F L46ds55AiY	A	raw hub p identical to the parent for shared compound-steps	True (128 compound-steps)
G Y72ky4006o	A	pathways at BH q < 0.05 (job; parent)	7; 6
G Y72ky4006o	A	pathways gained	KEGG: Metabolic pathways; KEGG: Nitrogen cycle
G Y72ky4006o	A	pathways lost	KEGG: Glucosinolate biosynthesis
G Y72ky4006o	A	Spearman rho of combined p	0.7697
G Y72ky4006o	A	hub compounds tested (job; parent)	32; 32
G Y72ky4006o	A	hubs at FDR < 0.05 (job; parent)	0; 11
G Y72ky4006o	A	hubs gained	none
G Y72ky4006o	A	hubs lost	(S)-Malate; 2-Isopropylmalate; 2-Oxoglutarate; Glucobrassicin; Glutathione; L-Ascorbate; L-Glutamate; L-Isoleucine; L-Leucine; L-Valine; gamma-L-Glutamyl-L-cysteine
G Y72ky4006o	A	raw hub p identical to the parent for shared compound-steps	False (128 compound-steps)
H N2XgxyWlJn	A	pathways at BH q < 0.05 (job; parent)	0; 6
H N2XgxyWlJn	A	pathways gained	none
H N2XgxyWlJn	A	pathways lost	KEGG: Glucosinolate biosynthesis; KEGG: Plant hormone signal transduction; KEGG: Protein processing in endoplasmic reticulum; MapMan: Biotic Stress; MapMan: Cellular response overview; MapMan: Large enzyme families overview
H N2XgxyWlJn	A	Spearman rho of combined p	0.6419
H N2XgxyWlJn	A	hub compounds tested (job; parent)	32; 32
H N2XgxyWlJn	A	hubs at FDR < 0.05 (job; parent)	0; 11
H N2XgxyWlJn	A	hubs gained	none
H N2XgxyWlJn	A	hubs lost	(S)-Malate; 2-Isopropylmalate; 2-Oxoglutarate; Glucobrassicin; Glutathione; L-Ascorbate; L-Glutamate; L-Isoleucine; L-Leucine; L-Valine; gamma-L-Glutamyl-L-cysteine
H N2XgxyWlJn	A	raw hub p identical to the parent for shared compound-steps	False (128 compound-steps)
I sD7V1nZ4t4	D	pathways at BH q < 0.05 (job; parent)	27; 25
I sD7V1nZ4t4	D	pathways gained	KEGG: Fatty acid biosynthesis; KEGG: Fatty acid degradation; KEGG: Fatty acid metabolism; KEGG: Sulfoquinovose metabolism; MapMan: Inositol Phosphates
I sD7V1nZ4t4	D	pathways lost	KEGG: Ascorbate and aldarate metabolism; KEGG: Pentose and glucuronate interconversions; MapMan: Secondary Metabolism
I sD7V1nZ4t4	D	Spearman rho of combined p	0.8968
I sD7V1nZ4t4	D	hub compounds tested (job; parent)	32; 32
I sD7V1nZ4t4	D	hubs at FDR < 0.05 (job; parent)	16; 21
I sD7V1nZ4t4	D	hubs gained	none
I sD7V1nZ4t4	D	hubs lost	2-Isopropylmalate; 2-Oxoglutarate; L-Arginine; L-Aspartate; L-Glutamate
I sD7V1nZ4t4	D	raw hub p identical to the parent for shared compound-steps	False (128 compound-steps)
J C3kVBAm3N5	D	pathways at BH q < 0.05 (job; parent)	5; 25
J C3kVBAm3N5	D	pathways gained	none
J C3kVBAm3N5	D	pathways lost	KEGG: Aminoacyl-tRNA biosynthesis; KEGG: Ascorbate and aldarate metabolism; KEGG: DNA replication; KEGG: Histidine metabolism; KEGG: Porphyrin metabolism; KEGG: Ribosome; KEGG: Starch and sucrose metabolism; KEGG: Valine, leucine and isoleucine degradation; MapMan: AA TCA; MapMan: ChloroPlast CustomArray CUSTOM MAPPING; MapMan: Chloroplast PlasTi-microarray pathway; MapMan: Large enzyme families overview; MapMan: Overview; MapMan: Prokaryotic Ribosome LSU assembly; MapMan: Ribosomal proteins overview; MapMan: Secondary Metabolism; MapMan: Seed custom arabidopsis Seed; MapMan: Sucrose-Starch; MapMan: Volatile Carotenoid; MapMan: photosynthesis
J C3kVBAm3N5	D	Spearman rho of combined p	0.606
J C3kVBAm3N5	D	hub compounds tested (job; parent)	32; 32
J C3kVBAm3N5	D	hubs at FDR < 0.05 (job; parent)	0; 21
J C3kVBAm3N5	D	hubs gained	none
J C3kVBAm3N5	D	hubs lost	(S)-Malate; 2-Isopropylmalate; 2-Oxoglutarate; Adenine; Adenosine; FAD; FMN; Guanosine; L-Arginine; L-Aspartate; L-Glutamate; L-Isoleucine; L-Leucine; L-Tryptophan; N-Acetyl-L-glutamate; NAD+; Nicotinamide; Pantothenate; Quercitrin; Riboflavin; UDP-alpha-D-galactose

(continued)

Job	vs	Quantity	Value
J C3kVBAm3N5	D	raw hub p identical to the parent for shared compound-steps	False (128 compound-steps)
K1 Y0V56yFW12	B	pathways at BH q < 0.05 (job; parent)	1; 24
K1 Y0V56yFW12	B	pathways gained	none
K1 Y0V56yFW12	B	pathways lost	KEGG: Biosynthesis of secondary metabolites; KEGG: Carbon fixation by Calvin cycle; KEGG: Carbon metabolism; KEGG: Fatty acid biosynthesis; KEGG: Fatty acid degradation; KEGG: Fatty acid metabolism; KEGG: Glycolysis / Gluconeogenesis; KEGG: Metabolic pathways; KEGG: Photosynthesis - antenna proteins; KEGG: Porphyrin metabolism; KEGG: Pyruvate metabolism; KEGG: Riboflavin metabolism; KEGG: Selenocompound metabolism; KEGG: alpha-Linolenic acid metabolism; MapMan: AGPs; MapMan: ChloroPlast CustomArray CUSTOM MAPPING; MapMan: Chloroplast PlasTi-microarray pathway; MapMan: Isopentenyl PP; MapMan: Pentose Phosphate; MapMan: Phylloquinone Synthesis; MapMan: Prokaryotic Ribosome LSU assembly; MapMan: Seed custom arabidopsis Seed; MapMan: photosynthesis
K1 Y0V56yFW12	B	Spearman rho of combined p	0.6372
K1 Y0V56yFW12	B	hub compounds tested (job; parent)	32; 32
K1 Y0V56yFW12	B	hubs at FDR < 0.05 (job; parent)	0; 17
K1 Y0V56yFW12	B	hubs gained	none
K1 Y0V56yFW12	B	hubs lost	(S)-Malate; 2-Isopropylmalate; 2-Oxoglutarate; Adenine; Adenosine; FAD; FMN; Guanosine; L-Aspartate; L-Glutamate; L-Isoleucine; L-Leucine; L-Tryptophan; L-Valine; NAD+; Pantothenate; Riboflavin
K1 Y0V56yFW12	B	raw hub p identical to the parent for shared compound-steps	False (128 compound-steps)
K2 7725j3CkOp	B	pathways at BH q < 0.05 (job; parent)	0; 24
K2 7725j3CkOp	B	pathways gained	none
K2 7725j3CkOp	B	pathways lost	KEGG: Biosynthesis of cofactors; KEGG: Biosynthesis of secondary metabolites; KEGG: Carbon fixation by Calvin cycle; KEGG: Carbon metabolism; KEGG: Fatty acid biosynthesis; KEGG: Fatty acid degradation; KEGG: Fatty acid metabolism; KEGG: Glycolysis / Gluconeogenesis; KEGG: Metabolic pathways; KEGG: Photosynthesis - antenna proteins; KEGG: Porphyrin metabolism; KEGG: Pyruvate metabolism; KEGG: Riboflavin metabolism; KEGG: Selenocompound metabolism; KEGG: alpha-Linolenic acid metabolism; MapMan: AGPs; MapMan: ChloroPlast CustomArray CUSTOM MAPPING; MapMan: Chloroplast PlasTi-microarray pathway; MapMan: Isopentenyl PP; MapMan: Pentose Phosphate; MapMan: Phylloquinone Synthesis; MapMan: Prokaryotic Ribosome LSU assembly; MapMan: Seed custom arabidopsis Seed; MapMan: photosynthesis
K2 7725j3CkOp	B	Spearman rho of combined p	0.6466
K2 7725j3CkOp	B	hub compounds tested (job; parent)	32; 32
K2 7725j3CkOp	B	hubs at FDR < 0.05 (job; parent)	0; 17
K2 7725j3CkOp	B	hubs gained	none
K2 7725j3CkOp	B	hubs lost	(S)-Malate; 2-Isopropylmalate; 2-Oxoglutarate; Adenine; Adenosine; FAD; FMN; Guanosine; L-Aspartate; L-Glutamate; L-Isoleucine; L-Leucine; L-Tryptophan; L-Valine; NAD+; Pantothenate; Riboflavin
K2 7725j3CkOp	B	raw hub p identical to the parent for shared compound-steps	False (128 compound-steps)
K1 Y0V56yFW12	C	hubs at FDR < 0.05 (job; C)	0; 5
K1 Y0V56yFW12	C	hubs shared with C	none
K1 Y0V56yFW12	C	pathways at BH q < 0.05 (job; C)	1; 10
K2 7725j3CkOp	C	hubs at FDR < 0.05 (job; C)	0; 5
K2 7725j3CkOp	C	hubs shared with C	none
K2 7725j3CkOp	C	pathways at BH q < 0.05 (job; C)	0; 10
A 6kBOS7AN5K		pathways with a Gene expression p	232
A 6kBOS7AN5K		Spearman rho of combined p with the Gene expression p	0.7093
A 6kBOS7AN5K		pathways with a Proteomics p	223
A 6kBOS7AN5K		Spearman rho of combined p with the Proteomics p	0.594
A 6kBOS7AN5K		pathways with a Metabolomics p	80
A 6kBOS7AN5K		Spearman rho of combined p with the Metabolomics p	0.3922
A 6kBOS7AN5K		relevant compounds (robust) in the layer	45
A 6kBOS7AN5K		relevant compounds hub-tested	32
A 6kBOS7AN5K		relevant compounds with no hub result	2-Methylpropyl glucosinolate; 3-Amino-3-(4-hydroxyphenyl)propanoate; 6-Hydroxynicotinate; D-Sedoheptulose; Glucoalyssin; Glucoraphanin; Indolelactate; Lactulose; Loganin; Naringin; O-Acetyl-L-homoserine; Purine; Rutin
B JfVZ1L37PY		relevant compounds (robust) in the layer	45
B JfVZ1L37PY		relevant compounds hub-tested	32
B JfVZ1L37PY		relevant compounds with no hub result	2-Methylpropyl glucosinolate; 3-Amino-3-(4-hydroxyphenyl)propanoate; 6-Hydroxynicotinate; D-Sedoheptulose; Glucoalyssin; Glucoraphanin; Indolelactate; Lactulose; Loganin; Naringin; O-Acetyl-L-homoserine; Purine; Rutin
C wgT3IPi7U2		relevant compounds (robust) in the layer	45
C wgT3IPi7U2		relevant compounds hub-tested	32
C wgT3IPi7U2		relevant compounds with no hub result	2-Methylpropyl glucosinolate; 3-Amino-3-(4-hydroxyphenyl)propanoate; 6-Hydroxynicotinate; D-Sedoheptulose; Glucoalyssin; Glucoraphanin; Indolelactate; Lactulose; Loganin; Naringin; O-Acetyl-L-homoserine; Purine; Rutin
D Es6j23q40o		relevant compounds (robust) in the layer	45

(continued)

Job	vs	Quantity	Value
D Es6j23q40o		relevant compounds hub-tested	32
D Es6j23q40o		relevant compounds with no hub result	2-Methylpropyl glucosinolate; 3-Amino-3-(4-hydroxyphenyl)propanoate; 6-Hydroxynicotinate; D-Sedoheptulose; Glucoalyssin; Glucoraphanin; Indolelactate; Lactulose; Loganin; Naringin; O-Acetyl-L-homoserine; Purine; Rutin
B JfVZ1L37PY		hubs at FDR < 0.05: raw p range	0.0007856-0.00729
B JfVZ1L37PY		hubs at FDR < 0.05: distinct adjusted p values	0.04441

Table S11: **Metabolite hubs (T05) and the lit–dark overlap (T18).** Raw p / BH FDR (best step) per job: A 6kBOS7AN5K, B JfVZ1L37PY, C wgT3IPi7U2, D Es6j23q40o; – not tested. Ties in FDR follow from the BH step-up. Overlap and reaction-count tests: hypergeometric over compounds hub-tested in both: $p = 0.91$; hub labels shuffled within KEGG reaction-count tertiles: $p = 0.93$; job A: Mann-Whitney, hubs vs other hub-tested compounds: $p = 0.83$; job B: Mann-Whitney, hubs vs other hub-tested compounds: $p = 0.10$; job C: Mann-Whitney, hubs vs other hub-tested compounds: $p = 0.29$; job D: Mann-Whitney, hubs vs other hub-tested compounds: $p = 0.11$.

Compound	KEGG	A	B (light)	C (dark)	D
L-Glutamate	C00025	2.3×10^{-4} / 0.0053 (3)	0.0050 / 0.0444 (4)	0.1115 / 0.3747 (4)	0.0105 / 0.0447 (4)
L-Leucine	C00123	2.7×10^{-4} / 0.0053 (4)	0.0073 / 0.0444 (4)	0.0184 / 0.1507 (3)	0.0133 / 0.0473 (2)
L-Valine	C00183	2.7×10^{-4} / 0.0053 (4)	0.0073 / 0.0444 (4)	0.0184 / 0.1507 (3)	0.0195 / 0.0657 (2)
L-Isoleucine	C00407	2.7×10^{-4} / 0.0053 (4)	0.0073 / 0.0444 (4)	0.0184 / 0.1507 (3)	0.0133 / 0.0473 (2)
gamma-L-Glutamyl-L-cysteine	C00669	2.9×10^{-4} / 0.0053 (4)	0.1118 / 0.2805 (4)	0.0959 / 0.3719 (4)	0.1699 / 0.3397 (3)
2-Isopropylmalate	C02504	1.5×10^{-4} / 0.0053 (4)	7.9×10^{-4} / 0.0444 (3)	5.8×10^{-4} / 0.0192 (4)	0.0101 / 0.0445 (3)
L-Ascorbate	C00072	7.9×10^{-5} / 0.0053 (1)	0.2537 / 0.4919 (2)	0.4471 / 0.8187 (4)	0.1489 / 0.3123 (4)
Glutathione	C00051	4.8×10^{-4} / 0.0068 (3)	0.3027 / 0.5224 (4)	0.2175 / 0.5673 (3)	0.1752 / 0.3397 (3)
(S)-Malate	C00149	9.2×10^{-4} / 0.0107 (3)	0.0020 / 0.0444 (2)	0.0014 / 0.0362 (3)	0.0038 / 0.0204 (3)
Glucobrassicin	C05837	0.0029 / 0.0248 (3)	0.4846 / 0.7053 (4)	0.3569 / 0.8001 (3)	0.0759 / 0.1905 (1)
2-Oxoglutarate	C00026	0.0041 / 0.0292 (4)	0.0039 / 0.0444 (4)	0.0060 / 0.1010 (2)	0.0091 / 0.0416 (4)
N-Acetyl-L-glutamate	C00624	0.0092 / 0.0550 (4)	0.0485 / 0.1477 (3)	0.0272 / 0.1661 (3)	0.0056 / 0.0288 (3)
L-Aspartate	C00049	0.0105 / 0.0587 (4)	0.0048 / 0.0444 (4)	0.3630 / 0.8001 (3)	0.0091 / 0.0416 (4)
L-Tyrosine	C00082	0.0120 / 0.0638 (3)	0.0100 / 0.0511 (2)	0.1431 / 0.4163 (2)	0.0227 / 0.0728 (4)
5-Oxoproline	C01879	0.0139 / 0.0686 (4)	0.1631 / 0.3538 (4)	0.1145 / 0.3747 (4)	0.2041 / 0.3579 (4)
1-O-Sinapoyl-beta-D-glucose	C01175	0.0164 / 0.0773 (4)	0.4849 / 0.7053 (4)	0.6154 / 0.8656 (3)	0.2147 / 0.3637 (4)
L-Tryptophan	C00078	0.0296 / 0.1185 (4)	0.0028 / 0.0444 (3)	0.1060 / 0.3747 (3)	0.0066 / 0.0325 (3)
NAD+	C00003	0.1518 / 0.4317 (4)	0.0047 / 0.0444 (4)	6.0×10^{-4} / 0.0192 (4)	7.4×10^{-4} / 0.0068 (2)
D-Gluconic acid	C00257	0.1800 / 0.4716 (2)	0.0218 / 0.0799 (4)	0.4365 / 0.8187 (3)	0.6707 / 0.8863 (4)
Riboflavin	C00255	0.2593 / 0.6036 (1)	0.0045 / 0.0444 (3)	0.1200 / 0.3747 (4)	3.2×10^{-4} / 0.0041 (3)
Pantothenate	C00864	0.2835 / 0.6357 (2)	0.0011 / 0.0444 (4)	0.0562 / 0.2768 (4)	0.0011 / 0.0074 (3)
UDP-alpha-D-galactose	C00052	0.2965 / 0.6433 (3)	0.0988 / 0.2529 (4)	0.0063 / 0.1010 (4)	1.2×10^{-4} / 0.0019 (4)
Quercitrin	C01750	0.3255 / 0.6830 (4)	0.8690 / 0.9615 (2)	0.2064 / 0.5620 (1)	0.0038 / 0.0204 (3)
FMN	C00061	0.4968 / 0.9217 (1)	0.0066 / 0.0444 (3)	1.2×10^{-4} / 0.0079 (4)	3.3×10^{-5} / 8.0×10^{-4} (4)
Adenosine	C00212	0.5391 / 0.9585 (4)	0.0018 / 0.0444 (4)	0.0078 / 0.1112 (4)	3.4×10^{-6} / 4.4×10^{-4} (3)
Guanosine	C00387	0.6677 / 1.0000 (4)	0.0041 / 0.0444 (4)	0.0161 / 0.1507 (4)	1.3×10^{-5} / 8.0×10^{-4} (3)
Nicotinamide	C00153	0.8780 / 1.0000 (4)	0.0103 / 0.0511 (4)	0.0879 / 0.3663 (4)	5.2×10^{-4} / 0.0052 (4)
3-Deoxy-D-manno-octulosonate	C01187	1.0000 / 1.0000 (4)	0.7822 / 0.9030 (4)	1.0000 / 1.0000 (4)	1.0000 / 1.0000 (4)
FAD	C00016	0.6396 / 1.0000 (4)	0.0058 / 0.0444 (3)	1.2×10^{-4} / 0.0079 (4)	3.3×10^{-5} / 8.0×10^{-4} (4)
L-Arginine	C00062	0.8419 / 1.0000 (3)	0.0395 / 0.1274 (4)	0.2013 / 0.5601 (4)	0.0125 / 0.0473 (4)
Allantoate	C00499	0.8325 / 1.0000 (3)	0.0244 / 0.0869 (4)	1.0000 / 1.0000 (4)	0.3942 / 0.5669 (1)
Adenine	C00147	0.7776 / 1.0000 (4)	0.0040 / 0.0444 (4)	0.0272 / 0.1661 (4)	3.0×10^{-5} / 8.0×10^{-4} (4)
D-Glucose	C00031	–	–	–	–
L-Glutamine	C00064	–	–	–	–
Sucrose	C00089	–	–	–	–
D-Fructose	C00095	–	–	–	–
L-Asparagine	C00152	–	–	–	–
3-Phospho-D-glycerate	C00197	–	–	–	–
Maltose	C00208	–	–	–	–
beta-D-Glucose	C00221	–	–	–	–
alpha-D-Glucose	C00267	–	–	–	–
Starch	C00369	–	–	–	–
D-Ribulose 1,5-bisphosphate	C01182	–	–	–	–

Table S12: **OSD-38 proteome checks (T23).** limma moderated t for the membrane sets next to the pre-registered Student t (T12), the membrane–soluble partition check, the truncation and filter accounting, and OSDR's file listing (2 November 2026).

Check	Item	n	Result
membrane set, limma moderated t	BCAA degradation	17	z −0.504; q 0.595; T12 z −0.435; T12 observed none
membrane set, limma moderated t	Glyoxylate/dicarboxylate	41	z −3.368; q 0.001; T12 z −2.794; T12 observed down
membrane set, limma moderated t	Carbon fixation	38	z −3.790; q 5.0×10^{-4}; T12 z −3.904; T12 observed down

(continued)

Check	Item	n	Result
membrane set, limma moderated t	Starch and sucrose	28	z −2.480; q 0.015; T12 z −2.631; T12 observed down
membrane set, limma moderated t	Glycolysis	60	z −4.280; q 5.0×10^{-4}; T12 z −4.193; T12 observed down
membrane, limma vs Student t	Spearman of protein t	2,000	rho 0.992
partition	proteins quantified in both fractions: Spearman of log2FC	366	rho −0.080
partition	ath00630: members in both fractions	27	membrane −0.10; soluble 0.09
partition	ath00710: members in both fractions	30	membrane −0.10; soluble 0.03
partition	ath00500: members in both fractions	12	membrane −0.23; soluble 0.09
partition	ath00010: members in both fractions	35	membrane −0.05; soluble 0.04
soluble members	ath00630: members quantified in the soluble fraction	30	soluble 0.09
soluble members	ath00710: members quantified in the soluble fraction	33	soluble 0.03
soluble members	ath00500: members quantified in the soluble fraction	13	soluble 0.12
soluble members	ath00010: members quantified in the soluble fraction	36	soluble 0.05
accounting	data rows in the OSDR table	65,535	
accounting	last protein group present (cut inside it)	438	
accounting	protein groups kept (1-437, complete)	437	
accounting	AGI loci in the kept groups	433	
accounting	loci that appear in more than one kept group	4	
accounting	loci after the score, channel and two-peptide filters	433	
accounting	data rows in the OSDR table	65,535	
accounting	rows in block 1 (used)	48,799	
accounting	protein groups in block 1	2,828	
accounting	highest protein group number in block 1	2,948	
accounting	AGI loci in block 1	2,827	
accounting	loci after the score, channel and two-peptide filters	2,000	
OSDR file listing	(default): GLDS-38_soluble_proteome_WyattLab_soluble.csv, 9792292 bytes, created 2017-09-21, updated ”	295	
OSDR file listing	(default): GLDS-38_soluble_proteome_WyattLab_soluble.xlsx, 6900438 bytes, created 2017-09-21, updated ”	295	
OSDR file listing	(default): GLDS-38_membrane_proteome_WyattLab_membrane.csv, 9323992 bytes, created 2017-09-21, updated ”	295	
OSDR file listing	(default): GLDS-38_membrane_proteome_WyattLab_membrane.xlsx, 7235696 bytes, created 2017-09-21, updated ”	295	
OSDR file listing	?all_files=true: GLDS-38_soluble_proteome_WyattLab_soluble.csv 9792292 bytes, created 2017-09-21, update	295	
OSDR file listing	?all_files=true: GLDS-38_soluble_proteome_WyattLab_soluble.xlsx 6900438 bytes, created 2017-09-21, updat	295	
OSDR file listing	?all_files=true: GLDS-38_membrane_proteome_WyattLab_membrane.csv, 9323992 bytes, created 2017-09-21, upda	295	
OSDR file listing	?all_files=true: GLDS-38_membrane_proteome_WyattLab_membrane.xlsx, 7235696 bytes, created 2017-09-21, upd	295	

Table S13: **Hypoxia positive control (T24) and signature addendum (T25).** OSD-136: Ws roots, 50 kPa without against with O_2 restored to 21 kPa (GeneLab limma t). GSE14502: Col-0 (35S line) shoots and roots, 2 h hypoxia, total mRNA (limma). A set is hypoxia-sensitive if up at $q < 0.05$ in a GSE14502 tissue; the BRIC comparator result is informative only for such sets. The addendum signature is the genes up (down) more than two-fold at $q < 0.05$ in both GSE14502 tissues. Gene rows: q is GeneLab's or limma's adjusted p.

Dataset	Set	Role	z	q	Outcome	t
OSD-136	Hypoxia-responsive	positive control	−0.5	0.8492	none	
OSD-136	Fermentation	positive control	9.6	9.0×10^{-4}	up	
OSD-136	DIN	positive control	0.0	0.9969	none	
GSE14502_shoot	Hypoxia-responsive	positive control	0.0	0.9969	none	
GSE14502_shoot	Fermentation	positive control	8.6	9.0×10^{-4}	up	
GSE14502_shoot	DIN	positive control	0.8	0.6344	none	
GSE14502_root	Hypoxia-responsive	positive control	0.0	0.9969	none	
GSE14502_root	Fermentation	positive control	6.8	0.0054	up	
GSE14502_root	DIN	positive control	2.9	0.0301	up	
OSD-38 BRIC-PDFU vs closed container (T12)	DIN	informative	−1.5	0.1646	none	
OSD-38 BRIC-PDFU vs closed container (T12)	Fermentation	informative	0.9	0.4504	none	
OSD-38 BRIC-PDFU vs closed container (T12)	Hypoxia-responsive	not informative	−0.3	0.7576	none	

(continued)

Dataset	Set	Role	z	q	Outcome	n
OSD-136	Hypoxia-up signature	addendum	22.2	3.0×10^{-4}	up	106
OSD-136	Hypoxia-down signature	addendum	−0.0	0.9748	none	10
OSD-38 BRIC-PDFU vs closed container	Hypoxia-up signature	addendum	13.0	3.0×10^{-4}	up	109
OSD-38 BRIC-PDFU vs closed container	Hypoxia-down signature	addendum	−2.9	0.0060	down	10
OSD-38 BRIC-PDFU vs closed container	ADH1	gene	$\log_2$FC 2.76	2.1×10^{-5}		
OSD-38 BRIC-PDFU vs closed container	PDC1	gene	$\log_2$FC 2.65	2.8×10^{-5}		
OSD-38 flight vs ground	Hypoxia-up signature	addendum	−3.2	0.0020	down	109
OSD-38 flight vs ground	Hypoxia-down signature	addendum	0.6	0.6130	none	10
OSD-38 flight vs ground	ADH1	gene	$\log_2$FC −0.28	0.7750		
OSD-38 flight vs ground	PDC1	gene	$\log_2$FC −0.23	0.8156		
GSE14502_shoot	ADH1	gene	$\log_2$FC 4.00	2.8×10^{-4}		
GSE14502_shoot	PDC1	gene	$\log_2$FC 6.12	2.2×10^{-4}		
GSE14502_root	ADH1	gene	$\log_2$FC 8.34	2.8×10^{-7}		
GSE14502_root	PDC1	gene	$\log_2$FC 8.21	1.7×10^{-5}		

Table S14: **MetaboLights variance sources (T26)**, from each study's ISA sample and assay files. Groups / samples (per group). "Max 100": compounds whose maximum over the study is exactly 100, i.e. values reported as percent of maximum.

Study	Variant	Tissue	Groups used	Groups / n	Platform	Max 100	Other organisms
MTBLS22	Arabidopsis thaliana.Col-0	rosette leaf	Source Name rg1 (0 mM NaCl)	1 / 6 (6)	Agilent 6890N GC; LECO Pegasus III, time-of-flight	68 of 68	none
MTBLS41	Arabidopsis thaliana.Col-0	rosette leaf	Source Name rg1 (0 mM NaCl)	1 / 5 (5)	Agilent 6890N GC; LECO Pegasus III, time-of-flight	67 of 67	none
MTBLS42	Arabidopsis thaliana.Col-0	rosette leaf	Source Name rg1 (0 mM NaCl)	1 / 6 (6)	Agilent 6890N GC; LECO Pegasus III, time-of-flight	67 of 67	none
MTBLS2289	Col-0	shoot	Genotype Col-0 WT, Drought stress water, shoot; one group per timepoint	7 / 35 (5)	Agilent 6890N GC; LECO Pegasus III, time-of-flight	0 of 75	none
MTBLS4066	Col-0	leaf	Sample group primary_sample_wildtype	1 / 6 (6)	Agilent 6890N GC; LECO Pegasus III, time-of-flight	0 of 50	none
MTBLS363	Arabidopsis thaliana str. Col-0	rosette leaf	Arabidopsis assay only (the study also profiles Agave americana); one group per elapsed time	8 / 24 (3)	Agilent 7890A GC; Agilent 5975C inert XL MSD, quadrupole	0 of 77	Agave americana

References

[1] Zhu, F. *et al.* The natural variance of Arabidopsis secondary metabolism on extended darkness. *Scientific Data* **11**, 841 (2024).

[2] Zhu, F. *et al.* Genome-wide association of the metabolic shifts underpinning dark-induced senescence in Arabidopsis. *The Plant Cell* **34**, 557–578 (2022).

[3] Wu, S. *et al.* Mapping the Arabidopsis Metabolic Landscape by Untargeted Metabolomics at Different Environmental Conditions. *Molecular Plant* **11**, 118–134 (2018).

[4] Caldana, C. *et al.* High-density kinetic analysis of the metabolomic and transcriptomic response of Arabidopsis to eight environmental conditions. *The Plant Journal* **67**, 869–884 (2011).

[5] Transcription profiling by array of *Arabidopsis* after exposure to different temperatures and light levels. ArrayExpress, E-MTAB-375 (2011). https://www.ebi.ac.uk/biostudies/arrayexpress/studies/E-MTAB-375.

[6] Proteomics and Transcriptomics analysis of *Arabidopsis* Seedlings in Microgravity. NASA Open Science Data Repository, OSD-38 (BRIC-20) (2015). https://osdr.nasa.gov/bio/repo/data/studies/OSD-38.

[7] Dissecting Low Atmospheric Pressure Stress: Transcriptome Responses to the Components of Hypobaria in *Arabidopsis* [Experiment 2]. NASA Open Science Data Repository, OSD-136 (GEO GSE87868) (2017). https://osdr.nasa.gov/bio/repo/data/studies/OSD-136.

[8] Zhou, M. *et al.* Dissecting Low Atmospheric Pressure Stress: Transcriptome Responses to the Components of Hypobaria in Arabidopsis. *Frontiers in Plant Science* **8**, 528 (2017).

[9] Immunopurified mRNA-ribosome complexes expose cell-type specific plasticity in response to hypoxia in *Arabidopsis*. NCBI Gene Expression Omnibus, GSE14502 (2009). https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE14502.

[10] Mustroph, A. *et al.* Profiling translatomes of discrete cell populations resolves altered cellular priorities during hypoxia in *Arabidopsis*. *Proceedings of the National Academy of Sciences* **106**, 18843–18848 (2009).